Compact mixed-kernel equations with actual Miller-pairing controls

Rational graph plus orthogonal p-eigenspace
All inputs are public curve and point data

Complete Miller and Frobenius constraints
P-192:23678 variables,95695 quartic equations

Actual extension-field kernel replay
Six targets,49 kernel points each; quotient evaluation remains open

Derived construction. For prime-order P-192 with Q=xG!=O, define the orthogonal Frobenius block W_p by [r]U=[r]V=O, pi(U)=[p]U, pi(V)=[p]V, and e_r(G,U)e_r(Q,V)=1. Nondegeneracy of the pairing between the two Frobenius eigenspaces makes W_p a line of r geometric points. Combining it with the rational graph Gamma=<(G,Q)> gives H=Gamma+W_p, the mixed maximal isotropic kernel of order r^2 from round51. All defining circuit coefficients lie in F_p. No explicit non-rational eigenpoint or target logarithm is needed to compile the equations. Preconditions include p>3, odd prime r!=p, p!=1 mod r, and nonzero G,Q generating the same rational order-r subgroup. [pcc:unverified]

What is constructed. compile.py emits a deterministic stream of actual polynomial expressions, records its SHA256, variables, output wires, equation count and degree bound, and supports exporting the stream. It uses the complete 44-equation addition predicate of round52, first-order slope/inverse witnesses, multiplication gates, and Miller line functions. The full optimized P-192 presentation has 23678 variables and 95695 equations of degree at most four. Its projection is H as a finite geometric point set. A quotient abelian-surface evaluator and a P-192 elimination are not performed here. The universal projection argument is written, with explicit small-field controls. [pcc:unverified]

Miller gate. A complete addition gate gives S=A+B, slope s, and selectors I,J,C,D,F for the two identity branches, cancellation, doubling, and generic addition. Let R=(u,v,q) be the evaluation point in normalized coordinates, A=(x,y,o), and S=(X,Y,O). Introduce one g and impose q(g-1)=0; (1-q)(I+J)(g-1)=0; (1-q)C(g-u+x)=0; (1-q)(D+F)[g(u-X)-v+y+s(u-x)]=0. These four equations have degree at most four. For nonzero evaluation arguments in the opposite Frobenius eigenspace, all required line denominators and vertical-line values are nonzero. The group-law branches include cancellation at the final r-th multiple. [pcc:unverified]

Zero arguments and signs. When either Miller source or evaluation point is the identity, the circuit assigns the artificial value one. Define the actual pairing to be one if either argument is zero; otherwise it is -f_(r,P)(R)/f_(r,R)(P), since r is odd. The complete orthogonality equation is (2o_U-1)(2o_V-1) f_(r,G)(U) f_(r,Q)(V) - f_(r,U)(G) f_(r,V)(Q)=0. This is quartic. It includes (U,V)=(O,O) and correctly treats exactly one zero argument. Multiplication of denominators introduces no extra solutions on the torsion/eigenspace domain, where every nonzero-argument Miller factor is nonzero. [pcc:unverified]

Torsion and Frobenius constraints. Four binary Miller chains start at G,U,Q,V. Each group step introduces 10 variables and 44 equations, one line value with four equations, and one accumulator with one equation: 12 variables and 49 equations per step. Requiring their final point to be O supplies the r-torsion constraints. Two additional fixed-scalar chains compute [p mod r]U and [p mod r]V; reduction of the scalar is valid after imposing r-torsion. Four coordinate power circuits compute u_x^p,u_y^p,v_x^p,v_y^p. Their values and unchanged identity flags are equated to the scalar-chain endpoints. This describes the p-eigenspace over the algebraic closure without choosing an extension-field basis. [pcc:unverified]

Count audit. Put m_r=bitlength(r)+popcount(r)-2=333 and m_s=bitlength(p mod r)+popcount(p mod r)-2=141. The coordinate p-power chain has m_F=204 multiplication gates. The dual block has 6+48m_r+20m_s+4m_F=19626 variables and 27+196m_r+88m_s+4m_F=78519 equations. The constant 27 counts input-point equations, Miller endpoint equations, Frobenius equalities and the final pairing equation. Adding the round52 rational graph and two final additions gives 23678 variables and 95695 equations. These counts come from emitted constraints and an independent formula audit. [pcc:unverified]

Positive power chain. For p=2^192-2^64-1, build cached exponents 2^w-1 for w=1,2,4,8,16,32,64; concatenate widths 64,32,16,8,4,2,1 to obtain 2^127-1; square once, then 64 more times, then multiply by the cached 2^64-1 power. The resulting exponent is exactly p, with 204 multiplication gates and no inversion, so zero coordinates are included. The initial binary/full-p presentation used 29186 variables and 117523 equations. Its source, certificate and logs are preserved under attempt1. The optimized presentation changes both the eigenvalue scalar and the power chain; the exact bundle reduction is 5508 variables and 21828 equations. No runtime comparison follows from this count. [pcc:unverified]

Exact P-192 constraint counts from retained streams; two circuit substitutions; no timing data.

Actual extension-field construction. The control is E/F_11:y^2=x^3+2x+7, r=7, G=(6,2), with F_(11^3)=F_11[Z]/(Z^3+Z+4). The modulus has no root over F_11 and hence is irreducible. The producer obtains an actual nonzero p-eigenpoint T by curve arithmetic and verifies its order and Frobenius action. All six nonzero rational targets are supplied as points, without their logarithms. Actual Miller pairings test all 294 pairs from the p-eigenspaces and select seven per target; 42 accepted dual witnesses satisfy every compiled equation. A further 126 complete mixed-kernel witnesses include the rational graph and final output additions. Six nonorthogonal witnesses fail the final pairing equation, and 42 modified Miller accumulators are rejected. [pcc:unverified]

Independent evidence. A separate checker uses packed-integer field elements, polynomial long reduction, independent point arithmetic and recursive Miller evaluation. It checks 49 bilinearity controls, all 294 pairing membership decisions, the six 49-point mixed kernels, and 14406 closure/isotropy tests in the verified torsion basis. It also audits the power exponent and size formulas. Omitting the Frobenius restrictions admits the full pairing-orthogonal space of r^3 points, so the pairing equation alone is an explicit negative control. All checks pass locally. Compiler-witness equation checks and their counts are producer instrumentation; the separate checker independently checks field, point, pairing and kernel semantics. [pcc:unverified]

Export and scope. The full P-192 constraint stream was exported temporarily, counted and hashed against the compiler record, then removed as a regenerable derived file. export-verification.json records its byte count, SHA256 and replay command. The P-192 evidence is a compact symbolic kernel presentation; actual extension-point and full-witness tests use the declared small control. The algebra of the projected P-192 kernel has dimension r^2, and compact input size alone does not provide an efficient elimination or quotient map. [pcc:unverified]

Literature. Miller's original algorithm and straight-line representation are standard: https://crypto.stanford.edu/miller/ . The normalized Weil-pairing formula is documented by the Magma handbook: https://docs.magma-maths.org/ArithmeticGeometry/EllipticCurvesOverFiniteFields/pairings.html . The new work here is the explicit constraint composition, its domain handling, size audit and replayable control; no novelty claim is made for Miller evaluation or binary scalar circuits.

Next construction. Use the compact mixed-kernel presentation to construct and test quotient functions, retaining the marked action. Investigate whether CM operators permit trace/norm or invariant arithmetic without expanding a dimension-r^2 basis. The non-rational pairing block is now specified and implemented; the quotient evaluation and its cost are the next concrete dependency.